

CS 5600: Database Project Description

The project requires building a database application for a real-world scenario of your choices. You will write a one or half-page narrative description of the database application you propose to work with throughout the course.

- What is the purpose of the database? Why is it needed?
What should it do?
- Who are the users and what are their information needs?
- What are the problems that the system should solve?
- What input data is available to the database?
- What kind of information should be stored in the database?

Additional Requirement:

As part of the project, students must integrate a Large Language Model (LLM) feature into their database application. For example, the system should support a chatbot or question-and-answer functionality that allows users to interact with the database in natural language. This will help demonstrate the practical use of LLMs for enhancing user experience and accessibility.

Note.

You can submit the project description in the document files such as .doc, .docx, .pdf, ..., etc. and upload on the Brightspace.

For example: CS5600DESC_XXXXX_YYYYY.doc, where XXXXX is CRN and YYYYY is your group number.

Project Description Sample

Project Name: The Indiana University Herbarium Database System

Description:

The Indiana University Herbarium, the official Herbarium for the State of Indiana, is a research museum with over 140,000 specimens. It serves as the specimen repository for faculty and student researchers at Indiana University (IU). Since specimens are only loaned to established herbaria, not to individuals, a database application is essential for improving access to this valuable scientific resource.

Purpose of the Database:

The database will make herbarium specimen information more accessible for researchers, students, and educators. It will provide up-to-date taxonomic information, standardized species lists, and county-level distribution data. The system will also reduce manual errors in data entry, improve cross-referencing of synonymous plant names, and facilitate collaborative research.

Users and Their Information Needs:

- Researchers: Require accurate taxonomic data, synonyms, and downloadable datasets for ecological or biodiversity studies.
- Students: Need a searchable and easy-to-use system for class projects, plant identification, and independent study.
- Educators: Will use the database as a teaching aid in plant science and ecology courses.
- General Public: May explore the system for environmental awareness and plant-related information.

Problems to Be Solved:

- Limited accessibility due to the Herbarium's remote location.
- Difficulty in retrieving and cross-referencing historical plant taxonomy.
- Lack of a centralized, standardized digital repository for Indiana's vascular plant species.

Input Data for the Database:

- Electronic species lists of vascular plants of Indiana.
- County-based distribution data from the Biodiversity of North America Project (BONAP).
- Synonyms standardized common names, and references from published field guides.
- Metadata such as collectors, authors, and specimen details.

Stored Information:

- Taxonomic records (species, subspecies, varieties).
- Collection metadata (collector, date, location, specimen ID).
- Distribution data by county and region.
- Synonyms and standardized common names.
- User queries and interaction logs (for system improvement).

Key Functions and Operations:

- Data Entry & Update: Authorized staff can add, edit, or delete specimen records, taxonomic information, and collection.
- Search and Retrieval: Users can query the database by scientific name, common name, collector, date, or county distribution.
- Cross-Referencing: The system links synonymous scientific names and provides standardized common names.

- Reporting: Generate summary reports such as the number of species per county, invasive species lists, or species diversity by region.
- Export Functionality: Researchers can download taxonomic or distribution data in spreadsheet formats for further analysis.
- Access Control: Different user roles (researcher, student, public) with appropriate permissions.
- Natural Language Interaction (LLM): A chatbot/Q&A interface allows users to ask questions in plain English, such as “*Which plants are native to Monroe County?*” or “*List endangered plants in Indiana.*” The system translates queries into database operations and returns relevant results.

Additional Requirement – LLM Integration:

As part of this project, the system must include a Large Language Model (LLM)-based feature to improve usability. For example, the database will support a chatbot or Q&A interface that allows users to interact with the system using natural language. This feature will enhance accessibility for non-technical users and expand the system’s outreach.

Reference:

[IU News – Digital Herbarium Complete](#)